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Slide preparation methods and apparatus

Abstract

Various examples are provided related to slide preparation. In one example, a slide preparation apparatus includes a support surface, an upper surface opposite the support surface, and a sample tube holder. The upper surface can include a slide channel extending to a stop, a fiducial mark disposed in the slide channel, and a support bar extending across the slide channel. The sample tube holder can secure a sample tube in a vertical position adjacent to an end of the upper surface adjacent to the stop. When a syringe is supported by a sample tube positioned in the sample tube holder, a needle of the syringe can be aligned with the fiducial mark when it is seated in a guide groove of the support bar.

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Background/Summary

BACKGROUND

(1) Microscope slides are prepared by hand for examination under a microscope. Positioning of the sample on the slide is an important aspect for automated processing. Consistency of sample position can facilitate smearing, treatment and examination of liquid or other samples. This can be especially true for automated slide processing.

SUMMARY

(2) Aspects of the present disclosure are related to slide preparation. In one aspect, among others, a slide preparation apparatus comprises a support surface; an upper surface opposite the support surface; and a sample tube holder affixed to an end surface of the slide preparation apparatus. The upper surface can comprise a slide channel extending from a first end of the upper surface to a stop adjacent to a second end of the upper surface; a fiducial mark disposed in the slide channel, the fiducial marker located a defined distance from the stop; and a support bar extending across the slide channel, the support bar comprising a guide groove aligned with the fiducial marker. The sample tube holder can be configured to secure a sample tube in a vertical position adjacent to the second end of the upper surface.

(3) In one or more aspects, the fiducial mark, guide groove and sample tube holder can be linearly aligned along a length of the slide channel. The fiducial mark, guide groove and sample tube holder can be linearly aligned along a center of the slide channel. The stop can extend across a width of the slide channel. The width of the slide channel can correspond to a width of a slide. The slide channel can comprise sidewalls on opposite sides of the slide channels, the sidewalls configured to constrain a slide in alignment with the fiducial mark, guide groove and sample tube holder. The support bar can extend across the slide channel over the sidewalls, the support bar positioned at a height corresponding to a thickness of the slide. The guide groove can be on a side of the support bar adjacent to the fiducial mark. An upper surface of the support bar can comprise a rounded portion

extending across the side of the support bar adjacent to the fiducial mark.

(4) In various aspects, the fiducial mark can define a target location for deposition of a sample on a slide inserted in the slide channel. The fiducial mark can be visible through the slide inserted in the slide channel. The fiducial mark can comprise a depression formed in a bottom surface of the slide channel. The sample tube holder can comprise a cylindrical chamber sized to receive a sample tube. The sample tube can be a blood collection tube. The sample tube holder can comprise a structure configured to support the sample tube. The structure can extend inward from a side of the cylindrical chamber. The structure can be a tab extending inward to a center of the cylindrical chamber. The structure can extend across the cylindrical chamber. The sample tube holder can comprise at least one opening extending through a side of the cylindrical chamber providing visual access to the sample tube. In some aspects, a needle of a syringe supported by a sample tube positioned in the sample tube holder can be aligned with the fiducial mark when the needle is seated in the guide groove of the support bar.

(5) Other systems, methods, features, and advantages of the present disclosure will be or become apparent to one with skill in the art upon examination of the following drawings and detailed description. It is intended that all such additional systems, methods, features, and advantages be included within this description, be within the scope of the present disclosure, and be protected by the accompanying claims. In addition, all optional and preferred features and modifications of the described embodiments are usable in all aspects of the disclosure taught herein. Furthermore, the individual features of the dependent claims, as well as all optional and preferred features and modifications of the described embodiments are combinable and interchangeable with one another.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Many aspects of the present disclosure can be better understood with reference to the following drawings. The components in the drawings are not necessarily to scale, emphasis instead being placed upon clearly illustrating the principles of the present disclosure. Moreover, in the drawings, like reference numerals designate corresponding parts throughout the several views.

(2) FIGS. 1A-1C illustrate an example of a slide preparation apparatus, in accordance with various embodiments of the present disclosure.

(3) FIGS. 2A and 2B illustrate another example of a slide preparation apparatus, in accordance with various embodiments of the present disclosure.

(4) FIGS. 3A-3C illustrate an example of a syringe positioned on the slide preparation apparatus of FIGS. 2A and 2B for deposit of a sample on a slide, in accordance with various embodiments of the present disclosure.

DETAILED DESCRIPTION

(5) Disclosed herein are various examples related to slide preparation. Reference will now be made in detail to the description of the embodiments as illustrated in the drawings, wherein like reference numbers indicate like parts throughout the several views.

(6) Referring to FIGS. 1A-1C, shown is an example of a slide preparation apparatus **100** that can be used by a user for placement of a sample on a slide. FIGS. 1A and 1C are perspective views of the slide preparation device **100**, and FIG. 1B includes top and side views of the slide preparation device **100**. As illustrated in these views, the slide preparation apparatus **100** comprises a support surface **103** on the bottom, side surfaces **106**, end surfaces **109**, and an upper surface **112** configured to assist with preparation of slides. The support surface **103** is configured to provide a stable base for the apparatus **100**. The support surface **103** can include feet or pads distributed to support the slide preparation apparatus **100** when placed on a surface for use. As shown in FIG. 1A, feet can be located on corners of the support surface **103**, which can be extended to avoid tipping of

the apparatus **100** during use.

(7) Opposite the support surface **103** is the upper surface **112**, which is configured to receive a slide for preparation. The upper surface **112** can comprise a slide channel **115** extending from a first end of the upper surface **112** to a stop **118** adjacent to a second end of the upper surface **112**. The slide channel **115** can be sized to receive the slide and secure it in alignment with other features of the slide preparation apparatus **100**. For example, the width of the slide channel **115** corresponds to the width of the slide with tolerance allowing the slide to be inserted and removed from the slide channel **115**. The slide channel **115** comprises a bottom surface **121** and sidewalls **124** defining the channel **115**. The sidewalls **124** can extend along the entire length of the slide channel **115** (across the slide preparation apparatus **100**) or can extend along a portion of the length of the slide channel **115**, as shown in FIGS. 1A-1C.

(8) The slide channel **115** is open at one end of the slide preparation apparatus **100** allowing the slide to be positioned within the slide channel **115**. A stop **118** at the other end of the slide channel **115** limits insertion of the slide in the channel **115**. In some implementations, the length of the slide channel **115** from the stop **118** to the end of the slide preparation apparatus **100** may be less than a length of the slide. With the slide inserted in the slide channel **115** and engaged with the stop **118**, the end of the inserted slide may extend beyond the end surface **109a** of the apparatus **100**. The stop **118** can extend across the width of the slide channel **115** between the sidewalls **124** or can be less than the width of the channel **115**. For example, the stop **118** can comprise one or more tabs or shoulders extending upward from the bottom surface **121** that limit the movement of the slide through the slide channel **115**. The stop **118** can be located adjacent to an end of the slide preparation apparatus **100**. For example, the stop **118** can extend across the end of the slide channel **115** along the end surface **109b** of the apparatus **100**. In some implementations, the stop **118** position may be adjustable to account for different slide dimensions.

(9) The slide channel **115** is open along the top to allow access to the surface of the slide for deposition of a sample. A fiducial mark **127** can be provided in the channel to indicate where a sample should be positioned on a slide positioned in the slide channel **115**. The fiducial mark **127** can be a defined length from the stop **118** and can be substantially centered between the sidewalls **124** of the slide channel **115** to indicate a preferred sample placement location. The fiducial mark **127** can be visible to a user of the slide preparation apparatus **100** through a transparent or semi-transparent slide positioned in the slide channel **115**. The fiducial mark **127** can be formed in the bottom surface **121** of the slide channel **115** with one or more color, texture, geometry, etc. For example, the fiducial mark **127** can be a circular depression or other patterned depression in the bottom surface **121**. The fiducial mark **127** can be colored or tinted to increase the visibility of the fiducial mark **127**. In other implementations, the fiducial mark **127** can be printed or embossed on the bottom surface **121** with one or more color, texture, geometry, etc.

(10) A support bar **130** can extend across a portion of the slide channel **115** at a height that allows the slide to pass under the support bar **130**. The support bar **130** can extend between the sidewalls **124** of the slide channel **115** at a height of about the thickness of the slide with tolerance allowing the slide to be inserted and removed from the slide channel **115**. The support bar **130** can help hold the slide in position against the bottom surface **121** of the slide channel **115**. As shown in FIGS. 1A-1C, the support bar **130** can extend across a portion of the slide channel **115** between the fiducial mark **127** and the end of the slide channel **115**. The support bar **130** can comprise a first side adjacent to the fiducial mark **127** and a second side opposite the first side.

(11) The first side of the support bar **130** comprises a guide groove **133** that can be positioned with respect to the fiducial mark **127**. For example, the guide groove **133** can facilitate alignment of a syringe needle with the fiducial mark **127** for deposit of a sample on the slide. The top of the support bar **130** (opposite the slide channel **115**) can include a rounded portion as seen in the side view of FIG. 1B, which can aid in positioning of the needle over the fiducial mark **127**. The second side of the support bar **130** can be separated from the first side by a width that allows the stop **118**

to be accessible for visual confirmation of engagement of the slide with the stop **118**, or the second side can extend to the end of the slide channel **115**. In FIGS. **1A-1C**, the second side of the support bar **130** is curved to allow visual access to the stop **118**. In other embodiments, the support bar **130** can extend from the first side to the end of the slide channel **115**, where the second side can be aligned with the end surface **109b** of the slide preparation apparatus **100**.

(12) The end surfaces **109** and side surfaces **106** of the slide preparation apparatus **100** extend between the upper surface **112** and the support surface **103**. The end and side surfaces **109** and **106** can extend vertically and/or at an angle between the upper and support surfaces **112** and **103** depending on the surface configurations. As illustrated in FIG. **1A**, the support surface **103** can be larger than the upper surface **112** for stability, with the side surfaces **106** extending between the two at a fixed angle. The side and/or end surfaces **106/109** can include text, logos, designs, or other embellishments. For example, a product name can be formed or printed on one or more of the side surfaces **106** and/or end surfaces **109**.

(13) The end surfaces **109** can extend vertically between the upper and support surfaces. The end surface **109a** at the open end of the slide channel **115** (opposite the stop **118**) can include a cavity **136** that extends into the slide preparation apparatus **100** from the end surface **109a** below the slide channel **115**. The cavity **136** can provide space for gripping the slide during insertion into or removal from the slide channel **115**. As shown in FIG. **1B**, the width of the cavity **136** can be less than the width of the slide channel **115** so that support of the slide is provided over the full length of the slide channel **115**. The depth of the cavity **136** can be varied to provide adequate access to the end of the slide for gripping.

(14) The slide preparation apparatus **100** also includes a sample tube holder **139** affixed to an end surface **109b** adjacent to the stop **118** of the slide channel **115**. The sample tube holder **139** can be configured to secure a sample tube in a vertical position adjacent to the end of the slide channel **115** adjacent to the stop **118**. The sample tube holder **139** can be permanently affixed to the end surface **109a** or can be detachably attached to the end surface **109a** of the slide preparation assembly **100**. The sample tube holder **139** can comprise a cylindrical chamber sized to receive a sample tube. The sample tube can be a blood collection tube or other tube suitable for collection of blood, cytoplasm, or other fluids. As seen in the top view of FIG. **1B**, the sample tube holder **139** comprises a structure **142** that extends inward from a side of the cylindrical chamber to support the sample tube. The structure **142** can be a tab that extends toward or across the center of the cylindrical chamber or can be a bar, circular base or endcap that extends across the cylindrical chamber. The sample tube holder **139** can also include one or more openings extending through a side of the cylindrical chamber to provide visual access to the sample tube in the holder **139**.

(15) As illustrated by the dashed line in the top view of FIG. **1B**, the fiducial mark **127**, guide groove **133** and sample tube holder **139** can be linearly aligned along a length of the slide channel **115**. For example, the fiducial mark **127**, guide groove **133** and sample tube holder **139** can be linearly aligned along a center of the slide channel **115**. Such alignment can facilitate deposition of a sample on a slide in the slide channel **115** using a syringe as will be discussed. The slide preparation apparatus can be fabricated from plastics, polymers, metals, glass or other appropriate materials.

(16) Referring next to FIGS. **2A** and **2B**, shown is another example of a slide preparation apparatus **100**. As illustrated in FIGS. **2A** and **2B**, the stop **118** and/or sidewalls **124** of the slide channel **115** can include additional tabs or clips **145** that can aid in securing the slide in position in the slide channel **115**. In addition, the sample tube holder **139** can include tapered edges to facilitate insertion of the sample tube into the cylindrical chamber.

(17) FIGS. **3A-3C** illustrate the use of the slide preparation apparatus **100** of FIGS. **2A** and **2B** for placement of a sample on a slide. FIG. **3A** is a top view illustrating a slide **303** secured in the slide channel **115** of the slide preparation apparatus **100**. The slide **303** is inserted into the slide channel **115**, under the support bar **130**, until it engages with the stop **118**. The support bar **130** and tabs or

clips **145** on the stop **188** and sidewalls **124** hold the slide **303** in position in the slide channel **115**. With the slide **303** in this position, the fiducial mark **127** indicates the location where the specimen should be deposited on the slide **303**. The fiducial mark **127** can be seen through a transparent or semi-transparent slide **303** by the user to assist with placement of the sample. A syringe **306** containing the sample can be positioned as illustrated in FIGS. **3A-3C**. As seen in the side view of FIG. **3B**, the syringe **306** can be supported by a sample tube **309** inserted in the sample tube holder **139** and by the support bar **130**. The sample can be extracted from the sample tube **309** prior to being positioned on the sample tube **309** or may be extracted from another sample tube or source. The sample can be extracted from the sample tube **309** before the sample tube **309** is inserted into the sample tube holder **139** or can be extracted from the sample tube **309** while it is located in the sample tube holder **139**.

(18) With the syringe **306** positioned on the sample tube **309** and the syringe needle in the guide groove **133** of the support bar **130**, as shown in the end view of FIG. **3C**, the end of the needle is located over the fiducial mark **127**. The sample can be discharged from the syringe needle at the desired location on the slide **303**. The alignment of the sample tube holder **139**, guide groove **133** and fiducial mark **127** can significantly aid in the consistent placement of samples on slides **303** for processing. Once the sample has been deposited on the slide **303**, the sample may be smeared or otherwise treated on the surface of the slide **303**. The slide **303** with the sample can be removed from the slide preparation apparatus **100** by grasping the end of the slide **303** over the cavity **136** in the end surface **109a**. The user can insert a finger or thumb under the slide **303** to grasp the slide **303** and pull it out of the slide channel **115**.

(19) It should be emphasized that the above-described embodiments of the present disclosure are merely possible examples of implementations set forth for a clear understanding of the principles of the disclosure. Many variations and modifications may be made to the above-described embodiment(s) without departing substantially from the spirit and principles of the disclosure. All such modifications and variations are intended to be included herein within the scope of this disclosure and protected by the following claims.

(20) The term “substantially” is meant to permit deviations from the descriptive term that don't negatively impact the intended purpose. Descriptive terms are implicitly understood to be modified by the word substantially, even if the term is not explicitly modified by the word substantially.

(21) It should be noted that ratios, concentrations, amounts, and other numerical data may be expressed herein in a range format. It is to be understood that such a range format is used for convenience and brevity, and thus, should be interpreted in a flexible manner to include not only the numerical values explicitly recited as the limits of the range, but also to include all the individual numerical values or sub-ranges encompassed within that range as if each numerical value and sub-range is explicitly recited. To illustrate, a concentration range of “about 0.1% to about 5%” should be interpreted to include not only the explicitly recited concentration of about 0.1 wt % to about 5 wt %, but also include individual concentrations (e.g., 1%, 2%, 3%, and 4%) and the sub-ranges (e.g., 0.5%, 1.1%, 2.2%, 3.3%, and 4.4%) within the indicated range. The term “about” can include traditional rounding according to significant figures of numerical values. In addition, the phrase “about ‘x’ to ‘y’” includes “about ‘x’ to about ‘y’”.

Claims

1. A slide preparation apparatus, comprising: a base having an upper surface; a slide channel extending along the upper surface; a support bar having a first portion and a second portion, said bar extending across and slightly above the slide channel, the support bar including a guide groove notch, the guide groove notch partially separating the first and second portions of the support bar; and a sample tube holder separate from the support bar and removably secured to an end surface of the slide preparation apparatus, the sample tube holder configured to secure a sample tube in a

vertical position adjacent to the upper surface.

2. The slide preparation apparatus of claim 1, wherein the notch in the support bar is substantially V-shaped.
 3. The slide preparation apparatus of claim 1, wherein the support bar includes a curved portion opposite of the notch.
 4. The slide preparation apparatus of claim 1, further comprising a stop extending across a width of the slide channel.
 5. The slide preparation apparatus of claim 4, wherein the width of the slide channel corresponds to a width of a slide.
 6. The slide preparation apparatus of claim 1, wherein the slide channel comprises sidewalls on opposite sides of the slide channel, the sidewalls configured to constrain a slide therebetween.
 7. The slide preparation apparatus of claim 3, further comprising a gap having a fixed distance between the curved portion of the support bar and the sample tube holder.
 8. The slide preparation apparatus of claim 1, further comprising a fiducial mark, wherein the guide groove notch is on a side of the support bar adjacent to the fiducial mark.
 9. The slide preparation apparatus of claim 4, wherein the stop extends substantially or entirely across the width of the slide channel.
 10. The slide preparation apparatus of claim 8, wherein the fiducial mark defines a target location for deposition of a sample on a slide inserted in the slide channel.
 11. The slide preparation apparatus of claim 1, wherein a top of the sample tube holder is positioned below the support bar.
 12. The slide preparation apparatus of claim 1, wherein the notch partially surrounds an opening partially separating the first and second portions of the support bar, and the sample tube holder is detachably secured to the end surface of the slide preparation apparatus.
 13. The slide preparation apparatus of claim 1, wherein the sample tube holder comprises a cylindrical chamber sized to receive a sample tube.
 14. The slide preparation apparatus of claim 12, wherein the sample tube is a blood collection tube.
 15. The slide preparation apparatus of claim 4, wherein the sample tube holder extends between the stop and a base of the apparatus.
 16. The slide preparation apparatus of claim 4, further comprising a cavity at an end of the upper surface opposite the stop.
 17. The slide preparation apparatus of claim 13, further comprising a support structure extending across an amount of the cylindrical chamber.
 18. The slide preparation apparatus of claim 17, wherein the support structure extends about half the distance across the cylindrical chamber.
 19. The slide preparation apparatus of claim 13, wherein the sample tube holder comprises at least one opening extending through a side of the cylindrical chamber.
 20. The slide preparation apparatus of claim 1, further comprising a fiducial mark at the upper surface, wherein the guide groove notch is configured to receive a needle and align the needle with the fiducial mark.
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